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Beyond competition: dynamics of constructional interaction between HALFNESS evaluative constructions in Italian

<https://doi.org/10.1515/flin-2025-0116>

Received August 2, 2025; accepted March 28, 2026; published online May 6, 2026

Abstract: This paper explores the relationship between two Italian evaluative constructions, *mezzo X* and *semi-X*, both coming from the expression of half-quantity. The aim is to assess whether a competition-based perspective adequately accounts for their coexistence. To this end, we conduct both a synchronic analysis, focusing on lexical and semantic productivity to uncover differences in usage patterns, and a diachronic analysis, tracing the evolution of each construction's productivity to identify whether their interaction is characterized by exclusion or inclusion. Synchronically, the productivity of each construction does not appear to be constrained by the presence of the other. Rather, we find overlaps across their domains of application, with frequent violations of isomorphism. Diachronically, the constructions do not replace one another; instead, they show increasing overlap, measured by the growing number of shared fillers. From the eighteenth century onwards, *semi-X* begins to converge and to overlap with *mezzo X* in the selection of adjectival fillers. However, *mezzo X* does not seem to exhibit any decline in productivity, nor a shift in its domain of application. These findings suggest that, in the case of evaluative constructions, coexistence may be more effectively accounted for by complex models of interaction, encompassing also inclusive or neutral relationships.

Keywords: productivity; competition; approximation; rivalry; evaluative constructions; constructional interactions

1 Introduction

In this contribution, we explore the concept of constructional interaction, and in particular the notion of competition, between evaluative constructions. We preliminarily define constructional interaction in relation to the concept of biological interaction, i.e., the relationship between organisms sharing the same biological

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niche (and the effects that they have on each other). Thus, constructional interaction is a cover term encompassing all possible relationships between constructions that somehow occupy the same linguistic niche, where niche indicates a specific domain of application or use of a construction. We address this topic by examining the relationship of coexistence between two constructions expressing HALFNESS in Italian, namely *semi-X* and *mezzo X* ‘half X’, mainly focusing on their evaluative extensions. The former is a prefixal construction involving the neoclassical prefix *semi-* (Iacobini 2004; Micheli 2023), whose original meaning is ‘half, half part of’ (1a). The latter, instead, is an analytic construction, as it includes the free lexeme *mezzo* ‘half’, which, at least in its quantitative meaning, is employed as quantifier (1b).

- (1) a. *semicerchio* ‘half-circle’
 b. *mezza arancia* ‘half an orange’

Interestingly, the two constructions seem to have followed a similar path from the expression of half quantity to qualitative reduction: independent studies have shown that both *semi-X* and *mezzo X* have undergone a semantic shift from their original quantitative function to markers of approximation and attenuation (Benigni 2023; Iacobini 2004; Micheli 2023). For instance, in (2a) *semi-estivo* approximates the description of the climate by referring to the adjective *estivo* ‘summery’ and then pointing to a reduced degree of some of the qualities described by that adjective. Similarly, *mezzo negozio* (2b) describes a place that shows some features of a *negozio* ‘shop’, but that does not fully coincide with that description.

- (2) a. *La giornata più calda [...] quasi con*
 DET.DEF.F.SG day.SG more hot.F.SG almost with
un clima semi- estivo
 DET.INDEF.M.SG climate.SG half summery. M.SG
 ‘The hottest day [...], almost with a near-summery climate.’
 (from Masini and Micheli 2020: 394)
- b. *Questo posto pur essendo un caffè [...]*
 DEM.PROX.M.SG place.SG even-if be.GER DET.INDEF.M.SG café.SG
è anche un mezzo negozio
 be.PRS.IND.3SG also DET.INDEF.M.SG half.F.SG shop.SG
 ‘This place, while being a café [...], is also a sort of shop.’
 (from Benigni 2023: 18)

The occurrences in (2) stand as metaphoric extensions of the original meaning of HALFNESS, and can be defined as evaluative, in that they do not refer to a part of the modified element, but indicate a deviation from its standard meaning (Grandi 2002: 29; Grandi and Körtvélyessy 2015: 11): e.g., *mezzo negozio* does not refer to ‘half of a shop’ but is best interpreted as a ‘non-prototypical shop’.

The two constructions have only been studied separately, and the similarity in their meanings and the common path they followed has not been explicitly acknowledged in the literature. Nonetheless, in both of their meanings they seem to give rise to semantically related forms: in particular, QUANTITY *mezzo* X is commonly employed as a paraphrase for QUANTITY *semi*-X in lexicographic descriptions (3):

- (3) a. *semicerchio* ~ *mezzo cerchio* ‘half-circle’
 b. *semiluna* ~ *mezzaluna* ‘half-moon’

(GRADIT, De Mauro 2007)

Similarly, in their evaluative interpretation, the two constructions seem to give rise to a number of quasi-synonymous doublets¹ (4a-b); however, this relationship does not seem to apply with all the possible bases, and in some cases a substitution test yields unfelicitous results (4c):

- (4) a. *una porta semichiusa* ~ *una porta mezza chiusa* ‘a half-closed door’
 b. *un clima semiestivo* (~) *un clima mezza estivo* ‘a near-summery climate’
 c. *una repubblica semipresidenziale*/ **mezzo/a presidenziale* ‘a semi-presidential republic’

While the relationship between *semi*-X and *mezzo* X has been overlooked in previous studies, it seems a fruitful field of inquiry in the study of coexistence and competition among evaluatives. It has been widely acknowledged that analytic and synthetic strategies of evaluation coexist in various languages, sometimes expressing similar evaluative meanings (Grandi 2002: 249 et seq.; Merlini Barbaresi 2015: 40–41). This is not unexpected, since it is known that evaluative constructions violate the blocking rule and produce potentially competing forms, be they structurally similar or not (Grandi 2002: 152–154, 2015: 79–81; Grandi and Montermini 2006).

The question is thus how the coexistence of *semi*-X and *mezzo* X is managed in Italian. In fact, they seem to cover, at least partially, a common semantic space. This looks like a (partial) violation of the principle of isomorphism (Haiman 1980), which predicts a tendency to associate a single meaning with a single form in a one-to-one correspondence. Thus, we should expect the two constructions to compete for the same semantic niche, leading to the exclusion or marginalization of one of the two patterns, or to complementary distribution. However, research has shown the peculiar behavior of evaluative constructions when compared to

¹ We should keep in mind, however, that the interpretation of evaluatives is often heavily context-dependent (Merlini Barbaresi 2015; Prieto 2015). Thus, for instance, *un clima mezza estivo* (lit. ‘a half-summery climate’) (4b) can be considered as a paraphrase of *un clima semiestivo* only in some contexts, while in others it may trigger a divergent interpretation.

other areas of word-formation (Cartier and Huyghe 2021; Huyghe and Varvara 2023; Tovená and Kihm 2008) as often multiple constructions can be freely used to express the same meaning, leading to polymorphy (or “inclusive competition”, cf. Masini et al. 2021).

In this study we focus on the productivity of *semi*-X and *mezzo* X, as the notion of productivity has been explicitly linked in the literature to competition (Bauer 2001; Fernandez-Dominguez 2017; Plag 1999). To do so, we first define the concept of competition and its manifestation between evaluative constructions (Section 2). We then present two case studies, aimed at uncovering the distribution of the constructions and the dynamics that led to such distribution over time. First, we investigate the lexical and semantic productivity of *semi*- and *mezzo* in synchronic perspective to assess whether isomorphism or polymorphism best describe the distribution of HALFNESS constructions (Section 3), and then we present some data on the diachronic evolution of their productivity, to assess which kind of dynamics have taken place among them, namely competitive exclusion or inclusion (Section 4).

2 Framing competition between evaluative constructions

2.1 Defining competition

The notion of competition, or rivalry, has originated and is mainly applied in the study of word formation (Aronoff 1976; Aronoff 2016; Huyghe and Varvara 2023; Bagasheva et al. 2024), and, more marginally, in inflection (e.g., Carstairs-McCarthy 1994; Thornton 2011, 2019). However, under the theoretical drive of constructionist studies, it has been extended to syntax (e.g., Goldberg 2019; Zehentner 2019), and to relationships between constructions that straddle the analytic/synthetic boundaries (Masini 2019).

We can preliminarily define rivalry as the relationship that arises between affixes or constructions that are potentially capable of expressing the same semantics, and that thus “compete” with one another for the coverage of the same semantic niche (Huyghe and Varvara 2023). The need for a mechanism of competition between (partially specified) constructions characterized by a degree of semantic equivalence is linked to the general tendency toward isomorphism in natural languages. This principle, variably referred to (isomorphism, no synonymy; cf. Bolinger 1968;

Goldberg 1995; Haiman 1980), predicts a tendency toward the avoidance of synonyms within a language:

The first type [= isomorphism] [...] is that of a one-to-one correspondence between the signans and the signatum, whether this be a single word or a grammatical construct. [...]. It also serves as the unspoken basis for the commonly accepted axiom that no true synonyms exist, i.e. that different forms must have different meanings [...].

(Haiman 1980: 515–516)

The existence of competing constructions may lead to blocking, which is regarded as one of the ways in which the creation of synonyms is avoided (Aronoff 1976: 43; Rainer 1989, 2016). The mechanism of blocking can operate both at the lexical level and at the pattern level (Rainer 2016). At the lexical level, the existence of a lexically specified construction precludes the existence of its potential synonyms. At the pattern level, instead, blocking operates by preventing the application of a rule (that is, a schema), in a domain that is already covered by another productive rule (or schema). In other words, if we have two productive and semantically similar constructions, the creation of lexically specified types in a particular linguistic niche by means of one of them should preclude the use of the other construction in the same niche. By virtue of blocking, thus, two semi-specified, semantically similar constructions are expected to cover distinct lexical domains; in other words, they should not be fully synonymous, as they are not expected to occur with lexical material selected by the competing construction. The coexistence of the two constructions should therefore have consequences for the productivity of both, constraining it.

However, it is well known that blocking sometimes fails to occur (Gardani et al. 2019: 18–19). In these cases, competition is assumed to emerge, operating at both the level of abstract affixes/patterns and at the level of actual types. In the former case, two or more semantically similar, semi-specified patterns compete to be exploited within the same lexical domains (i.e., with the same fillers), while in the latter, the actual forms created from synonymous constructions combined with the same bases compete for the expression of the same referent (Aronoff 2016). In our contribution, we will primarily focus on the first level, paying only marginal attention to the second.

This view of the relationship between constructions, at least as formulated by Aronoff (2016, 2017), draws on Gause's biological *Axiom of Competitive Exclusion*: the coexistence of two constructions tapping into the same lexical domain is conceptualized as the coexistence of two species within the same ecological niche. The expected outcome is that the constructions/species will attempt to occupy the same niche and exclude competitors from it. In cases where the competition is resolved, the result at the level of affixes/constructions could be either the emergence of a complementary distribution, or the demise of one of the competitors (Aronoff 2016).

However, often rivalry is not resolved, and some overlaps between constructions may persist. Regardless of the outcome, this interaction between constructions is fundamentally negative in nature: maintaining the biological parallel, competition is a type of ecological interaction in which both competitors are “harmed” (Figure 1).

In linguistic terms, we could say that a construction’s productivity is restricted by the presence of a competitor, since the theoretical goal is the realization/preservation of isomorphism. However, as shown in Figure 1, research in biology and ecology maintains that competition is only one of the possible ways in which two species can interact (e.g., Odum 1983: 393 et seq.; Mougi and Kondoh 2012): for example, alongside negative interactions such as competition and predation, we also find positive interactions, such as commensalism, in which one species benefits from its relationship with the other without causing harm, and mutualism, where they both benefit from the interaction. Similarly to ecological sciences, it has also recently been argued in linguistics that competition is only one of several possible interactions between constructions, which can be complex and multifaceted (Desagulier 2022; Dukic and Palmer 2024; Zehentner 2019).

To provide a strict and operationalizable definition capable of distinguishing competition from other types of interaction, we may consider competition on the background of three theoretical types of interactions between constructions:

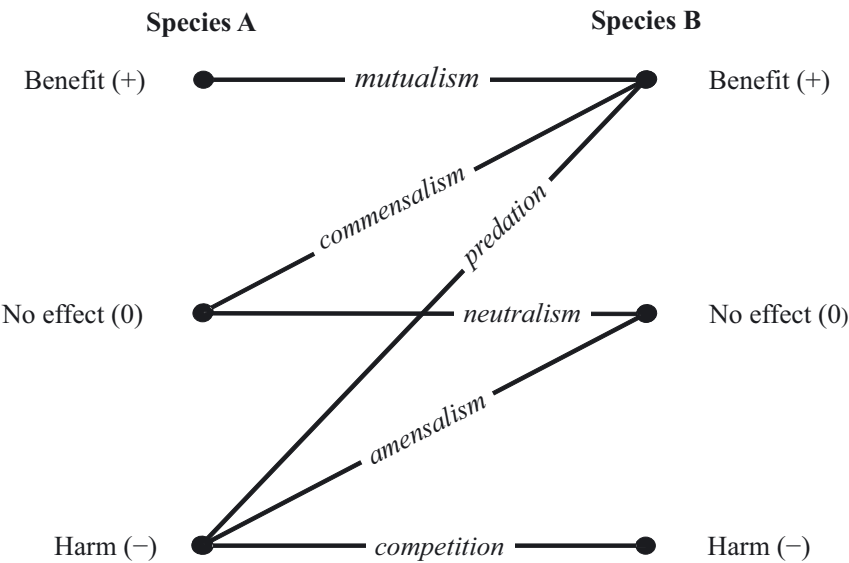


Figure 1: Taxonomy of the six main interspecific interactions, adapted from Le Roux et al. (2020: 388).

1. **Negative interaction:** growth in productivity of one construction hinders that of the other or drives it to occupy different lexical domains.
2. **Neutral/no interaction:** the two constructions remain in equilibrium, with no observable changes in productivity.
3. **Positive interaction:** the constructions mutually promote each other's growth in productivity, or one of the two promotes the other without being constrained in its productivity.

By considering such tripartition, we can define competition not as the simple coexistence between two quasi-synonymous constructions, but as a negative-signed interaction between them, aimed at the (partial) realization of isomorphism, and thus at the elimination or marginalization of one of the competitors.

2.2 Do evaluative constructions “compete” with one another?

The picture becomes more complex when examining the relationship between semantically similar constructions and affixes in evaluative morphology. First, it must be noted that the terms “competition” and “rivalry” are less frequently employed in relation to evaluative morphology than in studies on word-formation. As a matter of fact, there is a relatively limited number of studies that explicitly address competition between evaluatives. This seems counterintuitive, since one of the peculiarities of evaluative morphology is that it systematically violates the blocking rule (Dressler et al. 2019; Grandi 2002, 2015). This violation is interpreted by Grandi (2002: 153) as due to the fact that the semantic contribution of evaluative affixes is not referential, but rather relational in nature, and thus less cognitively relevant. In favour of this hypothesis is the fact that actual blocking is more commonly found when evaluative constructions get lexicalized and acquire a stable referential function (Grandi 2002: 153; Masini et al. 2023: 18). However, it is unclear what happens in all the other cases, and how the possibility of evaluating the same fillers with many different affixes/constructions is regulated at the system-level.

There are indeed some studies that approach the coexistence of semantically similar evaluative constructions by adopting a competition-based framework (Cartier and Huyghe 2021; Efthymiou 2024; Kamber and Huyghe 2023; Lacić 2025).² In

2 There are indeed other cases in which “affix rivalry” seems to be assumed simply on the basis of the fact that two constructions are not in complementary distribution (Dressler et al. 2019; Tarasova and Sánchez Fajardo 2024); in such cases, however, rivalry is probably understood as competition in context, that is, as the possibility of the online choice between multiple alternative affixes with the same base. This, however, represents a different level of competition from the one we have considered so far (competition as a system-level phenomenon).

some cases, it has been observed that there are collocation differences between evaluative constructions, and therefore areas of complementary distribution, along with areas of overlap between some of them (Efthymiou 2024); such overlaps can be quite substantial (cf., for example, Lacić 2025 on *kvazi(-)* and *nazovi(-)* in Croatian). In general, these studies report the presence of complex and non-univocal interactions between constructions; thus, relationships between evaluative constructions seem to go, more or less, in the direction of what Tovená and Kihm (2008: 19) refer to as “formal exuberance”, which in some cases does not appear to be motivated by relevant semantic distinctions between types.

Moreover, a purely synchronic perspective only shows a stage in a diachronic process of interaction, which could include overlaps, as well as niche differentiation between constructions; however, it does not show us the direction of the process, if there is one. On this note, studies on intensifying prefixes in French (Cartier and Huyghe 2021; Kamber and Huyghe 2023) have shown that not only complementary distribution between synonymous affixes and doublets is rarely found, but also that there are no signs of diachronic regression of one of the competitors; moreover, the development of new evaluative affixes is not used to cover different niches, but to intensify lexemes that are already employed with rival constructions (Kamber and Huyghe 2023: 79). This, in some cases, leads to the parallel growth in frequency of quasi-synonymous types created by means of different constructions (Cartier and Huyghe 2021: 19). This would suggest a relation of facilitation and mutual expansion of productivity, rather than one of rivalry.

Given the peculiar nature of rivalry among evaluative constructions, our question is whether the coexistence of *semi-X* and *mezzo X* is consistent with a competition model, or whether it is better accounted for by other possible types of constructional interactions. We will investigate this question by looking at the productivity of *semi-X* and *mezzo X* in synchrony and then focusing on how the productivity of these two constructions has evolved.

3 Isomorphism or polymorphism? The productivity of HALFNESS constructions

As we mentioned, productivity is one of the factors that is affected by the presence of competitors at the system level. In particular, we expect one of the competitors to be more productive than the other, or at least to observe differences in productivity between the competitors, corresponding to different domains of application (Baayen and Lieber 1991: 818; Plag 1999: 118). In this sense, productivity examined from a synchronic perspective is not an indisputable indication of whether competition is

taking (or has taken) place, but rather a possible symptom. Nevertheless, it can offer preliminary insights into the distribution of the two HALFNESS constructions.

To explore this aspect, we adopt a dual notion of productivity that takes into account both lexical productivity (productivity in the narrow sense), understood as the number of possible new types licensed by the constructions under investigation, and semantic productivity, which concerns the extent to which such constructions may be applied to fillers from the same semantic fields.

3.1 Data

The dataset was built based on the written standard Italian corpus CORIS (165 million words) (Rossini Favretti et al. 2002). Given the lack of morphological annotation in the corpus, and the strong polysemy of *mezzo* (which is also a noun, meaning ‘means, medium, vehicle’), an initial query was performed to identify a list of frequent words and patterns to be excluded from the search, as they formally contained the substrings *semi* and *mezzo* but did not correspond to actual types of the constructions under investigation³ (e.g., *seminario* ‘seminar’, [*a*] *mezzo posta* ‘by post’). Subsequently, the actual query was performed. The query included all the unbound and hyphenated occurrences of the strings *semi* and *mezzo* followed by another word, as well as bound occurrences (where one of the two strings is inserted at the beginning of a word). Considering the negative filter specified above, 31,065 total occurrences were retrieved. For feasibility reasons, we decided to analyze a subsample of the occurrences: thus, we made a random extraction, corresponding to 40 % of the total number. The resulting 12,426 occurrences were subsequently cleaned to exclude other false positives, reaching the final number of 11,246 total occurrences of types of *semi-X* (2,676 occurrences) and *mezzo X* (8,570 occurrences).

We then tagged the occurrences for the construction, the lemma and the part of speech of the filler, text genre, and additionally performed a broad semantic annotation, aimed at detecting the evaluative uses of the constructions. The initial criterion employed was the assessment of the presence/absence of quantification, to exclude occurrences with a quantitative function. However, given the exploratory nature of the annotation, several functional categories emerged.⁴ Therefore, to identify occurrences with an approximative or attenuative value, we checked whether the construction indicates a decrease of the property denoted by the filler,

³ The list of excluded elements can be found in the online repository, that will be made available upon acceptance.

⁴ The variety of uses of the two constructions will not be treated here, for reasons of space, and we will focus only on occurrences with an evaluative function. See Autore for a comprehensive treatment.

or the non-prototypicality of the referent indicated with respect to the category designated by the filler. Table 1 shows the results of the annotation of occurrences as approximative/attenuative or non-evaluative (*Other functions*), accompanied by the token count of the part of speech of the fillers. The annotation reveals that in the case of *semi-X*, evaluative occurrences are the most commonly found, while in the case of *mezzo X* evaluative occurrences constitute a minority of the sample.

As we see in Table 1, both *mezzo X* and *semi-X* turn out to be employed with fillers belonging to different lexical categories: adjectival (5a-6a), nominal (5b-6b), and even verbal (5c-6c), though mostly restricted to analytic forms including participles in the case of *mezzo X* (5c).

(5)	a.	[...] <i>un</i>	<i>pezzo</i>	<i>di</i>	<i>cartone</i>	<i>già</i>
		DET.INDEF.M.SG	piece.sg	of	cardboard.sg	already
		<i>mezzo</i>	<i>fradicio</i>	<i>di</i>	<i>pioggia</i>	
		half.M.SG	soaked.M.SG	of	rain.sg	
		'[...] a piece of cardboard already almost soaked with rain'				
						(CORIS, NARRAT)
b.	<i>I</i>	<i>virus</i>	<i>benigni [...]</i>	<i>finiranno</i>	<i>col</i>	
	DET.DEF.M.PL	virus.pl	benign.m.pl	end_up.fut.3sg	with-	
						DET.DEF.M.SG
	<i>produrre</i>	<i>una</i>	<i>mezza</i>	<i>rivoluzione</i>	<i>nel</i>	
	produce.inf	det.def.f.sg	half.f.sg	revolution.sg	in-DET.DEF.M.SG	
	<i>mondo</i>	<i>dell'</i>	<i>informatica</i>			
	world.sg	of-	computing.sg			
						DET.DEF.M.SG
		'Benign viruses will eventually bring about a partial revolution in the world of computing.'				
						(CORIS, PRACC)

Table 1: Semantic classification of the constructions involving *semi-X* and *mezzo X*, with token counts.

Form	Function	Part of speech (filler)				Total
		Noun	Adjective	Verb	Adverb	
<i>mezzo X</i>	Other functions	7,423	86	8	5	7,522
	ATTENUATION/APPROXIMATION	687	340	21	–	1,048
	Total	8,110	426	29	5	8,570
<i>semi-X</i>	Other functions	549	13	–	–	562
	ATTENUATION/APPROXIMATION	548	1,557	7	–	2,112
	Total	1,097	1,570	7	–	2,674

- c. *Le* *sale* *si* *erano* *già*
 DET.DEF.F.PL hall.PL REFL be.PST.IPFV.3SG already
mezze vuotate [...] half.F.PL empty.PTCP.PST.F.PL
 'The halls **had** already **half emptied**'
 (CORIS, NARRAT)
- (6) a. *Aveva* *un* *lato* *del*
 have.PST.IPFV.3SG DET.INDEF.M.SG side.SG of_DET.INDEF.M.SG
viso *blu* *e* *un*
 face.SG blue and DET.INDEF.M.SG
occhio *semichiuso*
 eye.SG semi-closed.M.SG
 'He had one side of his face blue and one eye **half-closed**'
 (CORIS, NARRAT)
- b. [...] *sette* *piani* *e* *diecimila*
 seven floor.SG and ten_thousand
metri *quadri* *in* *stato*
 meter.PL square.M.PL in state.PL
di *semiabbandono*
 of semi-abandonment.M.SG
 'Seven floors and ten thousand square meters in a state of **semi-abandonment**'
 (CORIS, STAMPA)
- c. *Nel* *maggio* '93 *un*
 in_DET.INDEF.M.SG May 1993 DET.INDEF.M.SG
violento *sisma* *semidistrusse* *la*
 violent.M.SG quake.SG semi-destroy.PST.PFV.3SG DET.DEF.F.SG
casa *che* *affittavano* [...] house.SG REL rent.PST.IPFV.3PL
 'In May '93 a violent earthquake **half-destroyed** the house they were renting'
 (CORIS, STAMPA)

Table 2 shows instead the distribution of the occurrences of *semi-X* and *mezzo X* across different text genres, highlighting that *mezzo X* is mostly employed in Fiction Prose, and to a slightly lesser extent in Press, while *semi-X* prevails in non-fiction genres, most notably in Academic and Legal Prose.

Table 2: Distribution of the occurrences by text genre.

	<i>mezzo X</i>	<i>semi-X</i>	Total
Blogs	27	94	121
Magazines	37	210	247
Fiction	596	759	1,355
Academic Prose	36	259	295
Legal Prose	0	160	160
Press	352	630	982
Total	1,048	2,112	3,160

3.2 Methods

3.2.1 Lexical productivity

As mentioned, we employ a twofold perspective on productivity. First, we assess the lexical productivity of *semi-X* and *mezzo X*, operationalized by employing quantitative measures of lexical diversity and openness to innovation (Baayen 2009). In particular, we look at the number of types (V) and hapaxes (V_1), as well as to three other measures:

- type/token ratio ($TTR = V/N$, where N stands for the number of tokens in the sample), which indicates the diversity of types of a construction;
- potential productivity, or hapax/token ratio ($P = V_1/N$), pointing at the potential for innovation of a construction;
- Shannon’s Entropy (H), another lexical diversity measure, which has been exploited and validated as a productivity measure (Lacić 2025; Pankratz et al. 2022). Entropy is calculated as:

$$(7) \quad H = - \sum_{v=1}^V \left(\frac{f_v}{N} \log_2 \left(\frac{f_v}{N} \right) \right)$$

where f_v corresponds to the token frequency of each type v .

We calculate all the measures in R by employing the *zipfR* package (Evert and Baroni 2007). Given the high sensitivity of the three employed measures to sample size (Gaeta and Ricca 2006), we calculate them on equal-sized samples. In order to avoid relying only on one random sample, we bootstrapped by replacement 1,000 samples of 500 occurrences for each construction, and calculated the mean value of each productivity measure across all iterations.

In order to get a fuller picture of the productivity of *semi-X* and *mezzo X*, we also look at their productivity in specific lexical categories, or parts of speech, and in specific text genres. By taking into account such aspects, we are able to get a

finer-grained picture of the coexistence between the two constructions. As for parts of speech, we assess the productivity of *mezzo* X and *semi*-X separately with nouns and adjectives, as these are the most frequent categories (Table 2). As for text genres, we decided to compare the productivity of *semi*-X and *mezzo* X, separately, in Fiction and in Press subcorpora from CORIS. We did not compare the two constructions in other genres, given the very low token frequency of *mezzo* X in such contexts (Table 2).

In both cases, we apply the same procedure as above, only on smaller samples: for each construction we bootstrap by replacement 150-tokens samples, calculate all the productivity measures, and re-iterate the procedure 1,000 times to get the mean value of the productivity rates.

3.2.2 Semantic productivity

As mentioned, we investigate productivity also from a semantic perspective. In particular, here we are interested in the semantic description of the domains of applicability of our constructions. While semantics represents only one of the possible dimensions in which niches can be identified (Rainer 2005), we decided to investigate the distribution of the constructions in semantic niches, as the semantic dimension has been deemed fundamental in determining productivity, especially in recent constructionist approaches to productivity.

For instance, high semantic openness, is often seen as a precondition for a construction's extensibility, and to correlate with its productivity (Suttle and Goldberg 2011); however, as noted by Van den Heede and Lauwers (2023), lexical and semantic productivity do not necessarily co-vary: a construction may accept many fillers while still being restricted to a narrow semantic niche. In this sense, discovering the semantic fields in which a construction can be productively employed can be seen as an assessment of some aspect of its availability (Corbin 1987): in our case, such an assessment suggests us whether *semi*-X and *mezzo* X are available for approximating fillers from the same semantic fields.

In order to operationalize semantic fields, we employ distributional representations of the fillers of the four constructions, following studies that have operationalized the impact of semantics on productivity (Hilpert 2016; Perek 2016; Van den Heede and Lauwers 2023). Distributional approaches to semantics are grounded in the distributional hypothesis (Firth 1957; Harris 1954), which states that words occurring in similar contexts tend to have similar meanings. In early distributional models, each word is represented as a vector derived from a co-occurrence matrix built from a reference corpus. In these so-called count-based models, each dimension of a word vector corresponds to a context word, and the values encode how often the target word co-occurs with that context within a given window. As a result, the

semantic profile of a word was explicitly defined by its distribution across observed contexts: words that occur in similar contexts are associated with similar vectors. Representing words as vectors allows them to be mapped as points in an n -dimensional space, where semantic similarity corresponds to spatial proximity: the closer two vectors are, the more semantically similar the corresponding words are.

While several previous studies on productivity and collocational behavior within Construction Grammar have sometimes relied on count-based distributional models (e.g. Hilpert 2016; Perek 2016), in the present study we follow more recent work in Construction Grammar in adopting predictive, neural approaches to distributional semantics (Budts and Petr   2020; Desagulier 2022). In fact, as noted in the literature, count-based models typically result in very high-dimensional and sparse representations, since the number of dimensions grows with the size of the vocabulary and most word–context combinations never occur (cf. the discussion in Desagulier 2022). Thus, in the present study, we adopt neural distributional embeddings trained with Word2Vec, specifically using the skip-gram architecture. Unlike count-based models, skip-gram does not directly store co-occurrence counts. Instead, it learns word representations by optimizing a predictive task: given a target word, the model is trained to predict its surrounding context words. Through this process, distributional information is compressed into a lower-dimensional latent space, whose dimensions are not directly interpretable but are learned so as to best capture regularities in word usage. The resulting vectors are dense and less affected by data sparsity.

To identify semantic fields within the set of fillers used by the constructions under investigation, we applied clustering techniques to their distributional representations. Clustering algorithms tend to group together items that occupy similar regions of the vector space and therefore share similar distributional properties. By clustering the fillers used by a construction, it becomes possible to assess the number of different fields in which a construction is exploited to create types.⁵ First, we extracted the set containing the lemmas of all fillers attested in both constructions. To obtain distributional representations for these lemmas, we used 200-dimensional vectors of the fillers calculated on the lemmatized version of the web-crawled Italian corpus itWaC (Baroni et al. 2009). The vectors were extracted by employing Word2Vec algorithm^{6,7} (Mikolov et al. 2013) using *gensim* (Rehurek and Sojka 2010).

5 Clearly, this should be not considered as a universally valid semantic classification, as it strongly depends on the onomasiological domain covered by the two constructions under scrutiny.

6 The model was trained with the skip-gram architecture on a window size ± 5 , for 15 epochs. I thank Nicola Cirillo for providing me with the trained model.

7 In 52 % of the cases of participial fillers (75 out of 142), no independent lemma was available in the model, as participles may be lemmatized as verbs. These fillers are equally distributed between *semi-X* and *mezzo X* (respectively, 45 and 40, 10 being shared by the two constructions). In such cases, we relied on the vector of the verb from which such participles are derived.

Then, we used UMAP algorithm from the *umap* package (McInnes et al. 2018) in Python to reduce the dimensionality of the vectors before clustering.⁸ We then applied k-means clustering to identify semantic fields. The optimal number of clusters (*k*) was determined by evaluating clusters for different values of *k* (from 2 to 15) employing the Davies-Bouldin index (Davies and Bouldin 1979) and the Calinski-Harabasz index (Calinski and Harabasz 1974). Based on these metrics,⁹ we judged the optimal number of clusters to be 12. To give a semantic description of the clusters found, we carried out a qualitative assessment of the words in each cluster, by giving priority to the most central words: in fact, for each cluster we extracted the words corresponding to the ten most central vectors (i.e., the ones closer to the centroid of the cluster). Since the most central vectors should correspond to the most representative words of each cluster, we prioritized these words in the description.

3.3 Results

3.3.1 Lexical productivity

Table 3 shows the number of types (*V*) and hapaxes (*V*₁) found for the two constructions in the full dataset. Instead, Figure 2 shows the mean productivity scores

Table 3: Number of tokens, types and hapaxes in the full dataset of *semi-X* and *mezzo X*.

	<i>mezzo X</i>	<i>semi-X</i>
<i>N</i>	1,048	2,096
<i>V</i>	398	450
<i>V</i> ₁	286	290

⁸ As noted by a reviewer, clustering on dimensionally reduced data is a debated practice: while it can help reduce noise in the data, by preserving aspects of the global structure in the case of UMAP, it may still distort distances and densities. Nonetheless, inspection of the most central fillers in the 12 UMAP clusters suggests semantically coherent groupings (cf. Table 6), as also confirmed by calculating the average distance of cluster members from the centroid (mean ≈ 0.54). Moreover, we also verified that clustering the original high-dimensional data leads to the same main finding: both constructions occupy all clusters, whether one chooses the optimal number of clusters for the non-reduced, high-dimensional data (2–4) or the 12 clusters found after UMAP. Therefore, employing dimensionally reduced data allows for a finer-grained and potentially more informative analysis, without altering the substantive results.

⁹ The results for *k* = 12 are 0.78 for Davies-Bouldin and 614.99 for Calinski-Harabasz. The table with the results from the evaluation task will be found in the online Appendix.

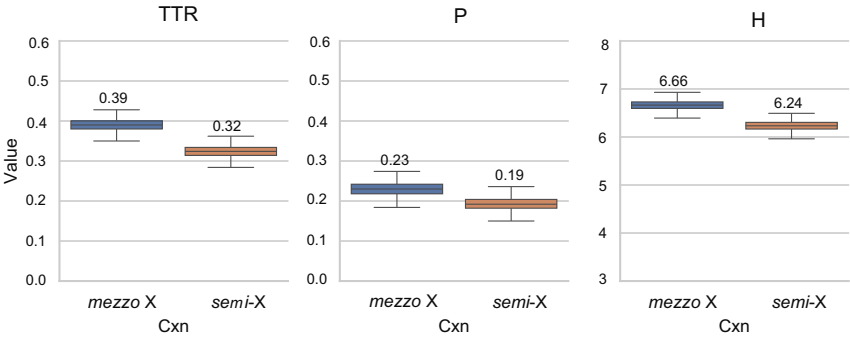


Figure 2: Productivity of HALFNESS constructions ($N = 500$).

calculated on samples of 500 tokens, as well as the distribution of the scores across all iterations.

Looking at Figure 2, we observe that *semi-X* and *mezzo X* display similar average rates of productivity. From the perspective of lexical diversity, *mezzo X* appears to be slightly more productive, particularly when considering *H*. This is likely due to the fact that, although there are some very frequent *mezzo X* types, the most frequent *semi-X* types (e.g. *seminterrato* ‘basement’, lit. ‘half-underground’; *semiautomatico* ‘semi-automatic’) have slightly higher token frequencies. For instance, the ten most frequent *mezzo X* types account for 37 % of the occurrences in the dataset, whereas the ten most frequent *semi-X* types account for a higher percentage, namely 44 %. Apart from these differences, the two constructions show a similar tendency towards innovation: as for *P*, they display quite similar values.

We further explore the difference in productivity between the two constructions by looking at their productivity with fillers from different parts of speech and in different text genres. Tables 4 and 5 show the counts for tokens, types and hapaxes, while Figures 3 and 4 show the productivity values on the 150 occurrences bootstrapped samples.

Table 4: Number of tokens, types and hapaxes of *semi-X* and *mezzo X*, by part of speech.

	Adjectives		Nouns	
	mezzo X	semi-X	mezzo X	semi-X
<i>N</i>	340	1,557	687	548
<i>V</i>	141	339	243	110
<i>V</i> ₁	98	214	175	69

Table 5: Number of tokens, types and hapaxes of *semi-X* and *mezzo X*, by text genre.

	Fiction		Press	
	<i>mezzo X</i>	<i>semi-X</i>	<i>mezzo X</i>	<i>semi-X</i>
<i>N</i>	596	759	352	630
<i>V</i>	245	183	146	198
<i>V</i> ₁	177	120	104	136

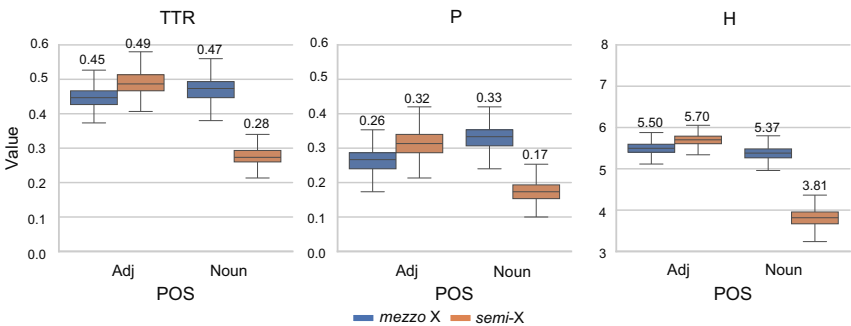


Figure 3: Productivity of HALFNESS constructions by part of speech (*N* = 150).

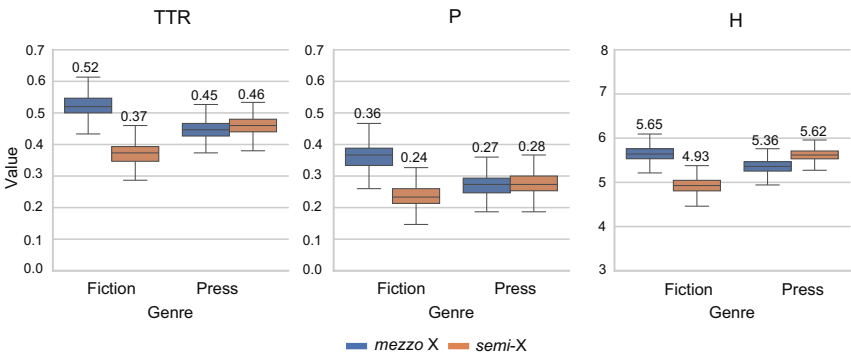


Figure 4: Productivity of HALFNESS constructions by text genre (*N* = 150).

As far as parts of speech are concerned, *mezzo X* is definitely more productive than *semi-X* in the with nominal fillers, showing an almost doubled level of lexical diversity. This seems consistent with the fact that *semi-X* generates far fewer nominal types (Table 4) and is more frequently used with adjectives; however, on the other hand, this result likely depends on our inclusion as nouns of three bases that can be

considered both nominal and adjectival, when found in a nominal construction. In fact, the relevant types (namely *seminterrato* ‘basement’, *semiautomatica* ‘semi-automatic gun’, *semiconduttore* ‘semi-conductor’) stand as top-ranking types in the distribution and account for more than half of the nominal uses of *semi-X* ($n = 295$). When removing them and recalculating the measures, *semi-X* still shows lower values than *mezzo X*, but much closer ones: lexical diversity becomes comparable ($TTR = 0.41$, $H = 5.21$), while the tendency to license hapaxes remains substantially lower ($P = 0.21$). By contrast *semi-X* appears to be slightly more productive than *mezzo X* with adjectives, although the distances are even smaller than in the nominal case, indicating very similar levels of lexical diversity and vitality for both constructions.

As for variation in productivity across text genres, *mezzo X* is used much more productively than *semi-X* in Fiction, showing a markedly higher level of lexical diversity as well as a greater tendency towards innovation. Conversely, in Press, the values for the two constructions are essentially the same, with only a slightly higher variation in H , indicating a marginally higher type diversity for *semi-X*.

3.3.2 Semantic productivity

Turning to semantic productivity, Figure 5 shows clusters obtained for the fillers of the two constructions, with the color coding showing whether a specific filler is employed with one of the two constructions, or both. Instead, in Table 6 we report our classification of the extracted clusters, along with their composition. Interestingly, *semi-X* and *mezzo X* cover all of the 12 available clusters, and they seem to be evenly distributed across clusters. In fact, in most of the clusters (7 out of 12), *semi-X* and *mezzo X* cover the semantic space to a largely comparable extent, as neither construction accounts for more than 65 % of the fillers within the cluster. In five clusters, however, we observe clear areas of divergence between the two constructions.

The clusters in which *mezzo X* is more strongly represented are prototypically characterized by event nouns. In the case of Cluster 3 (*urlo* ‘scream’), the fillers consist mostly of nouns denoting acts of emission (e.g. *risata* ‘laughter’, *singhiozzo* ‘sob’), but also bodily actions (e.g. *sorriso* ‘smile’, *occhiata* ‘glance’), often with an emotional connotation. In this cluster, *mezzo X* denotes an attenuated or weakened performance, or one that only partially fulfills the action denoted by the filler. A similar pattern is found in Cluster 10 (*scandalo* ‘scandal’), which is dominated by abstract nouns referring to negative outcomes (e.g. *fallimento* ‘failure’, *bancarotta* ‘bankruptcy’) or adverse events (e.g. *tragedia* ‘tragedy’). Here, *mezzo X* is used to downplay the negative impact of the event, mitigating the nominal semantics, especially when the lexical meaning of the noun is hyperbolic with respect to the actual event.

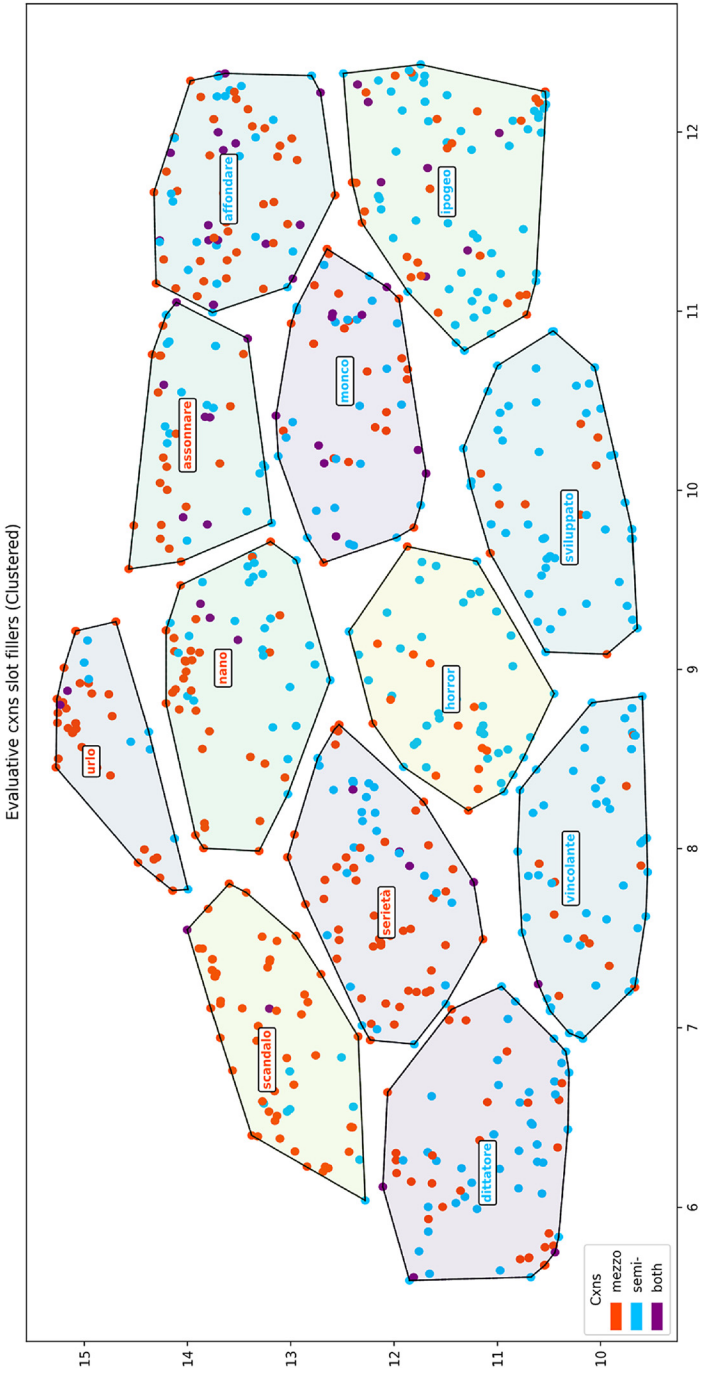


Figure 5: Clusters of semi-X and mezzo X fillers, labelled by the most central filler.

Table 6: Composition and description of the clusters found.

Cluster	Label	Most central words (<i>n</i> = 10)	Constructions	Total
0	Sociopolitical labels and concepts	<i>dittatore</i> ‘dictator’, <i>nazista</i> ‘Nazi’, <i>razzista</i> ‘racist’, <i>editto</i> ‘edict’, <i>forti</i> ‘strong’, <i>commissariato</i> ‘police station’, <i>fascista</i> ‘fascist’, <i>militarizzazione</i> ‘militarization’, <i>rivoluzione</i> ‘revolution’, <i>prussiana</i> ‘Prussian’	<i>semi-X</i> : 43 (58.11 %) <i>mezzo X</i> : 28 (37.84 %) both: 3 (4.05 %)	74
1	Physical and perceptual properties	<i>monco</i> ‘maimed’, <i>irriconoscibile</i> ‘unrecognizable’, <i>sbiadito</i> ‘faded’, <i>vacanza</i> ‘vacation’, <i>cartolina</i> ‘postcard’, <i>labirinto</i> ‘labyrinth’, <i>sbrindellato</i> ‘tattered’, <i>crepuscolare</i> ‘crepuscular’, <i>ombra</i> ‘shadow’, <i>diafano</i> ‘diaphanous’	<i>semi-X</i> : 25 (44.64 %) <i>mezzo X</i> : 21 (37.5 %) both: 10 (17.86 %)	56
2	Cognitive, behavioral and spiritual terms	<i>serietà</i> ‘seriousness’, <i>verità</i> ‘truth’, <i>segreto</i> ‘secret’, <i>zuffa</i> ‘brawl’, <i>ragione</i> ‘reason’, <i>certezza</i> ‘certainty’, <i>suo</i> ‘his’, <i>santo</i> ‘saint’, <i>vero</i> ‘true’, <i>desiderio</i> ‘desire’	<i>semi-X</i> : 26 (34.21 %) <i>mezzo X</i> : 46 (60.53 %) both: 4 (5.26 %)	76
3	Emission, performance and emotion-related events	<i>urlo</i> ‘scream’, <i>risata</i> ‘laughter’, <i>sbadiglio</i> ‘yawn’, <i>brivido</i> ‘shiver’, <i>ridere</i> ‘laugh’, <i>incubo</i> ‘nightmare’, <i>ringhio</i> ‘growl’, <i>singhiozzo</i> ‘hiccup’, <i>risolino</i> ‘chuckle’, <i>divertito</i> ‘amused’	<i>semi-X</i> : 8 (20 %) <i>mezzo X</i>: 30 (75 %) both: 2 (5 %)	40
4	Legal and political terms and procedures	<i>vincolante</i> ‘binding’, <i>direttiva</i> ‘directive’, <i>trasformazione</i> ‘transformation’, <i>intesa</i> ‘agreement’, <i>direttivo</i> ‘managerial’, <i>formalizzare</i> ‘formalize’, <i>materno</i> ‘maternal’, <i>processo</i> ‘process’, <i>obbligatorio</i> ‘mandatory’, <i>legge</i> ‘law’	<i>semi-X</i> : 46 (79.31 %) <i>mezzo X</i> : 11 (18.97 %) both: 1 (1.72 %)	58
5	Manufacture, industry and technology	<i>sviluppato</i> ‘developed’, <i>evoluto</i> ‘evolved’, <i>flessibile</i> ‘flexible’, <i>esaurimento</i> ‘exhaustion’, <i>potente</i> ‘powerful’, <i>attivo</i> ‘active’, <i>solido</i> ‘solid’, <i>intensivo</i> ‘intensive’, <i>standardizzare</i> ‘standardize’, <i>automatizzare</i> ‘automate’	<i>semi-X</i> : 49 (84.48 %) <i>mezzo X</i> : 9 (15.52 %) both: 0 (0 %)	58
6	Negative, object-related events and resulting states	<i>affondare</i> ‘sink’, <i>cadere</i> ‘fall’, <i>marcire</i> ‘rot’, <i>disfare</i> ‘undo’, <i>crollare</i> ‘collapse’, <i>sprofondare</i> ‘plunge’, <i>spopolare</i> ‘depopulate’, <i>rovesciare</i> ‘overturn’, <i>affogare</i> ‘drown’, <i>congelare</i> ‘freeze’	<i>semi-X</i> : 22 (30.14 %) <i>mezzo X</i> : 36 (49.32 %) both: 15 (20.55 %)	73
7	Body- and person-related properties and events	<i>assonnare</i> ‘make sleepy’, <i>russare</i> ‘snore’, <i>addormentato</i> ‘asleep’, <i>terrorizzato</i> ‘terrified’, <i>smarrito</i> ‘lost’, <i>accecare</i> ‘blind’	<i>semi-X</i> : 16 (36.36 %) <i>mezzo X</i> : 21	44

Table 6: (continued)

Cluster	Label	Most central words (n = 10)	Constructions	Total
8	Human attributes	<i>ipnotizzare</i> ‘hypnotize’, <i>dormiente</i> ‘dormant’, <i>svenire</i> ‘faint’, <i>stordito</i> ‘stunned’	(47.73 %) both: 7 (15.91 %)	65
		<i>nano</i> ‘dwarf’, <i>analfabeta</i> ‘illiterate’, <i>alcolista</i> ‘alcoholic’, <i>invalido</i> ‘disabled’, <i>mutolato</i> ‘mutilated’, <i>ladro</i> ‘thief’, <i>infermo</i> ‘sick’, <i>cieco</i> ‘blind’, <i>incoscienza</i> ‘unconscious’, <i>drogato</i> ‘drugged’	<i>semi-X</i> : 27 (41.54 %) <i>mezzo X</i> : 35 (53.85 %) both: 3 (4.62 %)	
9	Spatial and landscape-related terms	<i>ipogeo</i> ‘underground’, <i>sotterraneo</i> ‘subterranean’, <i>confortevole</i> ‘comfortable’, <i>abitabile</i> ‘habitable’, <i>bagnato</i> ‘wet’, <i>giardino</i> ‘garden’, <i>fiore</i> ‘flower’, <i>sempreverde</i> ‘evergreen’, <i>coperto</i> ‘covered’, <i>legnoso</i> ‘woody’	<i>semi-X</i> : 55 (62.5 %) <i>mezzo X</i> : 26 (29.55 %) both: 7 (7.95 %)	88
		<i>scandalo</i> ‘scandal’, <i>dramma</i> ‘drama’, <i>lutto</i> ‘mourning’, <i>farsa</i> ‘farce’, <i>sconfitta</i> ‘defeat’, <i>bancarotta</i> ‘bankruptcy’, <i>funerale</i> ‘funeral’, <i>fallimento</i> ‘failure’, <i>tragedia</i> ‘tragedy’, <i>crisi</i> ‘crisis’	<i>semi-X</i> : 8 (13.33 %) <i>mezzo X</i>: 50 (83.33 %) both: 2 (3.33 %)	
10	(Negative) event nouns	<i>horror</i> ‘horror’, <i>gotico</i> ‘gothic’, <i>olimpico</i> ‘Olympic’, <i>principiante</i> ‘beginner’, <i>musical</i> ‘musical’, <i>metaforico</i> ‘metaphorical’, <i>soprano</i> ‘soprano’, <i>casinò</i> ‘casino’, <i>professionistico</i> ‘professional’, <i>fantastico</i> ‘fantastic’	<i>semi-X</i> : 41 (74.55 %) <i>mezzo X</i> : 14 (25.45 %) both: 0 (0 %)	55

The clusters predominantly covered by *semi-X* also display common characteristics: they mainly contain adjectives belonging to specialized or technical lexicon, which appear to be particularly difficult to combine with *mezzo X*. Formally, these adjectives are often denominal, or more generally derived, and pertain to technical and specialized language. This is especially evident in Cluster 4 (*vincolante* ‘binding’) (8a) and Cluster 5 (*sviluppat*o ‘developed’) (8b), which include fillers related to the legal sphere and the productive/industrial one, respectively. In these clusters, *semi-X* is employed in technical language to create intermediate categories that do not fully correspond to the meaning of the referent denoted by the filler. This pattern can be observed, though to a lesser extent, also in Cluster 11 (*horror*) (8c), which contains terms related to the artistic and literary field. In this case too, *semi-X* is employed to create descriptive categories in art and literary criticism.

- (8) a. *palude*_N ‘swamp’ > *paludoso*_{Adj} ‘swampy’ > *semipaludoso* ‘semi-swampy’
 b. *monopolio*_N ‘monopoly’ > *monopolistico*_{Adj} ‘monopolistic’ > *semi-monopolistico* ‘semi-monopolistic’
 c. *biografia*_N ‘biography’ > *biografico*_{Adj} ‘biographical’ > *semibiografico* ‘semi-biographical’

Apart from these cases, the overlap between the two constructions seems to be quite consistent overall. Thus, the picture emerging from our analysis of the coexistence of *semi-X* and *mezzo X* reveals a nuanced situation: while there are broad areas of semantic overlap, there are also clusters in which the two constructions remain largely distinct. Overall, however, the two constructions appear to exhibit a substantial degree of overlap in the semantic fields encompassed by their domain of applicability.

3.3.3 Cases of shared productivity

An interesting phenomenon that emerges by looking at the color-coding in Figure 5 is the presence of a substantial number of fillers shared by the two constructions, namely 54, which are distributed over almost all of the clusters. Notably, shared fillers are mostly adjectival (38 out of 54): in particular, most of them are participial adjectives (*coperto* ‘covered’, *abbandonato* ‘abandoned’, *aperto* ‘open’), or underived adjectives (*nudo* ‘naked’, *lungo* ‘long’). The fact that *semi-X* and *mezzo X* show overlap particularly in non-technical, non-specialized lexicon is reflected in the text genres in which most overlap occurs: Fiction (*n. of types* = 33), followed by Press (*n. of types* = 10), and other non-fiction genres (*n. of types* = 9). It is true indeed that shared fillers do not grant the creation of synonymous forms: in (9a) *semilibertà* refers to a specific legal measure in the Italian penal system, while *mezza libertà* in (9b) only qualifies the autonomy of the population as ‘constrained, not full’.

- (9) a. *semilibertà* ‘semi-custody, semi-liberty’
 b. [...] *la formazione di uno Stato*
 DET.DEF.F.SG formation.SG of DET.INDEF.M.SG state.SG
 nuovo pur nella condizione di
 new.M.SG though in-DET.DEF.M.PL condition.SG of
 “*mezza libertà*” *concessa dai tedeschi*
 half.F.SG freedom.SG grant.PTCP.PST.F.SG by-DET.DEF.M.PL German.PL
 “[...] the formation of a new state, even under the condition of “half
 freedom” granted by the Germans”

(CORIS, STAMPA)

However, by carrying out a cursory semantic analysis of the types with shared fillers in the dataset, based on subjective judgements of the author, we found that 90 % of them (49 out of 54) are suited to express (almost) the same meaning, as shown in the examples in (10) and (11).

- (10) a. [...] *questo* *esperimento* *è* *stato*
 DEM.PROX.M.SG experiment.SG be.PRS.IND.3SG be.PTCP.PST.M.SG
un *semi-fallimento*
 DET.INDEF.M.SG semi-failure.SG
 ‘This experiment was a **partial failure**’ (CORIS, PRACC)
- b. [...] *l’* *investimento* *in* *titoli* [...] *è*
 DET.DEF.M.SG investment.SG in security.PL be.PRS.IND.3SG
stato *un* *mezzo* *fallimento*
 be.PTCP.PST.M.SG DET.INDEF.M.SG half.M.SG failure.SG
 ‘The investment in securities was a **partial failure**’ (CORIS, STAMPA)
- (11) a. [...] *un* *biglietto* [...] *che* *era* *già*
 DET.INDEF.M.SG ticket.SG REL be.PST.IPFV.3SG already
mezzo *cancellato*
 half.M.SG delete.PTCP.PST.M.SG
 ‘A note that was already **partly faded**’ (CORIS, EPHEM)
- b. [...] *le* *pietre* *grigie* [...] *con* *le*
 DET.DEF.F.PL stone.PL grey.F.PL with DET.DEF.F.PL
scritte *semi-cancellate*
 writing.PL semi-delete.PTCP.PST.M.SG
 ‘The grey stones with **partly faded** inscriptions on them’ (CORIS, NARRAT)

Such data suggest that *mezzo* X and *semi*-X do show some tendency to be employed with a common set of fillers, and they have the possibility of creating potentially synonymous linguistic forms. It is also interesting that this occurs in the same textual contexts, and this evidence discourages a strictly exclusivist interpretation of the distribution of the two constructions and their types.

3.4 Interim summary

Our results suggest a complex situation of coexistence between evaluative *semi*-X and *mezzo* X. On the one hand, there are indeed some differences between the two constructions. While with adjectival fillers *semi*-X and *mezzo* X display comparable levels of productivity, *mezzo* X appears to be more productively employed than *semi*-X with nouns, showing particularly high availability with abstract nouns. *Semi*-X has its own

niche of productivity too, namely complex adjectives, in particular in specialized and technical contexts of use. This is plausibly connected with differences in the context of productivity of the two constructions: *mezzo X* seems to be used more productively in fiction prose, plausibly due to a higher perceived expressive load, whereas it does not appear to be employed in academic or scientific prose. On the other hand, these differences are accompanied by clear evidence of a strong overlap between the two constructions, including frequent violations of isomorphism. As a matter of fact, the two constructions create mutually paraphrasable doublets especially (but not only) when employed with non-derived adjectives and participles. Furthermore, the fact that both constructions are effectively available in almost all of the fields occupied by the other creates the potential for the emergence of new near-synonymous formations, in addition to the doublets already attested. Taken together, these observations suggest that there is no strong complementary distribution between the two constructions, but rather a mixed coexistence, with areas of strong overlap in productivity alongside areas that are largely restricted to one of the two patterns.

The remaining question, however, is whether these areas of divergence have emerged as a result of a pressure towards differentiation, in line with a competitive exclusion-based perspective. This cannot be taken for granted. As already noted, synchronic data alone do not provide much insight into interactional dynamics but rather reveal the outcome of such dynamics. For this reason, we turn to diachronic data to assess whether the current productivity of *semi-X* and *mezzo X* results from competition, that is, from a negative interaction between constructions, or instead from different types of interaction.

4 Attraction or differentiation? The co-evolution of HALFNESS constructions

In this Section, we present some preliminary diachronic data on the productivity dynamics of *mezzo X* and *semi-X*. We do this in order to understand how the distribution found in Section 3 has developed. In fact, the presence of overlap between the two constructions may result from different underlying processes.

In order to provide a first assessment of the type of relationship between *semi-X* and *mezzo X*, we will evaluate the changes in productivity of the two constructions as well as the convergence or divergence of their productivity domains.

4.1 Data and methods

For our diachronic analysis we employed CODIT (Micheli 2022), the largest balanced diachronic corpus of written Italian (33Mw). The corpus comprises six text genres and includes a period ranging from the XIII century to 1947. It is organized in five periods:

- 1. XIII century–1375: 4,511,873 words
- 2. 1376–1532: 6,589,372 words
- 3. 1533–1691: 7,496,501 words
- 4. 1692–1840: 6,996,485 words
- 5. 1841–1940: 7,502,574 words.

We queried all the patterns containing the strings *semi* and *mezz** followed by another string (both hyphenated or not) or as substrings in complex words. The query yielded 12,520 occurrences. After cleaning the dataset from false positives, the actual occurrences of *semi*-X and *mezzo* X were found to be 4,854 (respectively, 4,011 for *mezzo* X and 843 for *semi*-X). We manually annotated the lemma of the filler, the part of speech of the filler, and the semantics of the occurrences in order to provide a distinction between evaluative and non-evaluative occurrences. In the process, the historical dictionary of Italian GDLI (*Grande Dizionario della Lingua Italiana*; Battaglia and Barberi Squarotti 1961–2002) was employed to disambiguate dubious cases. The distribution between evaluative and non-evaluative occurrences is shown in Table 7.

In our analysis, we concentrate on the dynamics in productivity of evaluative occurrences. Employing traditional measures of productivity is sometimes problematic in diachronic studies, given the potentially small number of occurrences and the variation in sample size (cf. Barðdal et al. to appear). For this reason, we decided to rely on a simpler measure of “realized productivity”, for each period, namely type

Table 7: Evaluative versus non-evaluative constructions in CODIT.

Construction	n. Tokens
<i>mezzo</i> X	4,011
ATTENUATION/APPROXIMATION	900
Other functions	3,111
<i>semi</i> -X	843
ATTENUATION/APPROXIMATION	407
Other functions	436

counts. In order to control for the variation in size of the subcorpora, corresponding to different periods, type counts were also normalized: we calculated the average of the number of tokens for all the periods, and then we normalized type frequencies of each period in proportion to such average number.

Since we are interested in the dynamics between constructions, and evolutions in the areas of overlap between them, for each period we counted the numbers of shared types, and, from the second period onwards, we counted for each construction how many of the new types (i.e., previously unseen ones) had been previously employed with the other construction (*attracted new types*) or have come to be employed simultaneously with the other construction in the same period (*joint new types*).

4.2 Results and discussion

4.2.1 Evolution of productivity

Figures 6 and 7 show, respectively, the number of tokens and the number of types of evaluative *semi-X* and *mezzo X*, by period. In both cases, we see an increase in frequency of the two constructions across time. In particular, when looking at token frequency, *semi-X* seems to progressively get closer to *mezzo X*, especially in the last period (1841–1947). However, to properly evaluate the evolution in productivity of the two constructions, we should rather look at the number of types they license, as token frequency constitutes a confounding factor in diachronic studies on productivity (Barðdal et al. to appear) (Figures 8 and 9). Let us look first at the evolution in productivity of *mezzo X* and *semi-X* separately.

Although evaluative *mezzo X* has been considered in the literature as an emerging construction (Masini and Micheli 2020: 396), its evaluative use with adjectives seems to be already productive in the first period (XIII cent.-1375): the most frequent type is *morto* ‘dead’ ($n = 6$), and most of the types correspond to participial adjectives¹⁰ (e.g., *cotto* ‘baked’, *tramutato* ‘transformed’). The most significant increase in productivity for *mezzo X* occurs in the second period (1376–1532), particularly with adjectives, where productivity more than doubles, compared to the previous period. Most of the new attested types are still participial adjectives, though some derived adjectives also appear (e.g., *luminoso* ‘bright’, *ombroso* ‘shady’). Afterwards, productivity continues to rise steadily, though at a more moderate pace, and, especially in the last period, the increase is more pronounced with nominal

¹⁰ Such a behavior seems to license the use of *mezzo X* with verbs, which is relatively infrequent but consistent across time, as verbs are mostly found in analytic forms involving participles.

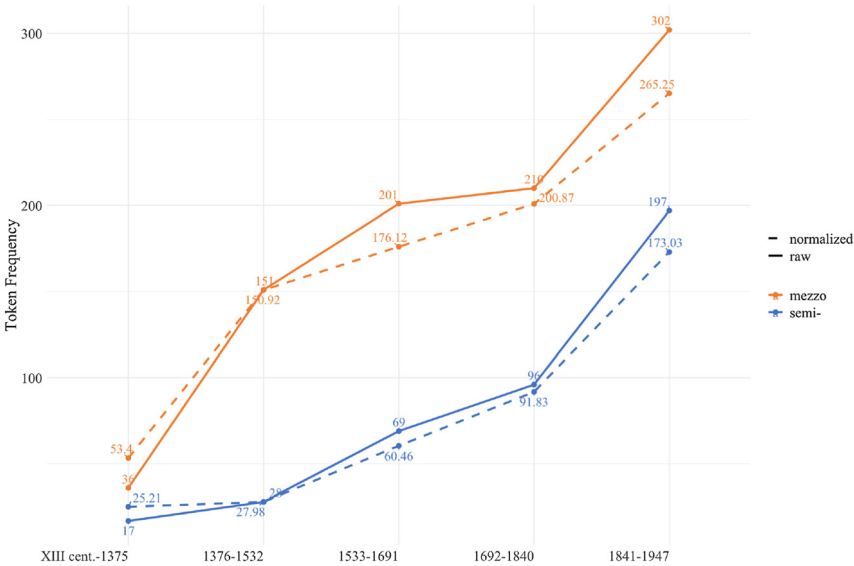


Figure 6: Token frequency (raw and normalized) of evaluative *semi-X* and *mezzo X*, by period.

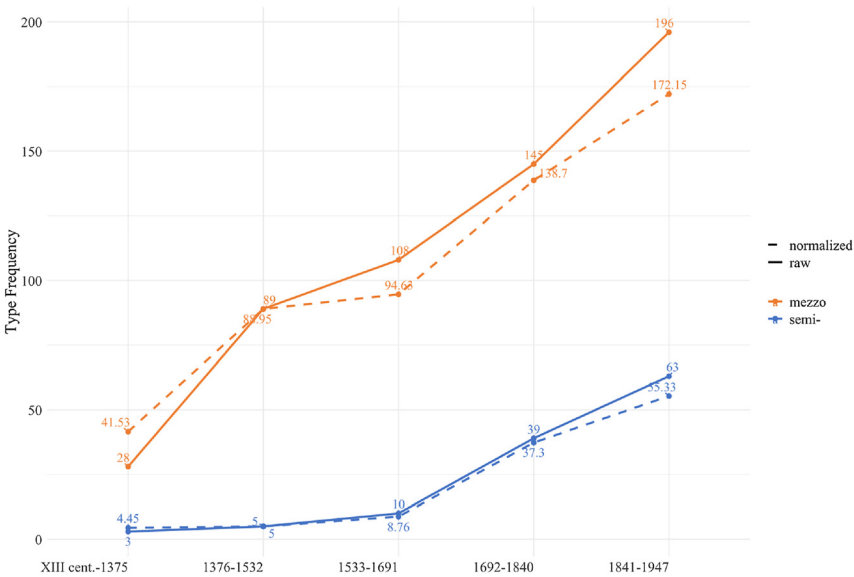


Figure 7: Type frequency (raw and normalized) of evaluative *semi-X* and *mezzo X*, by period.

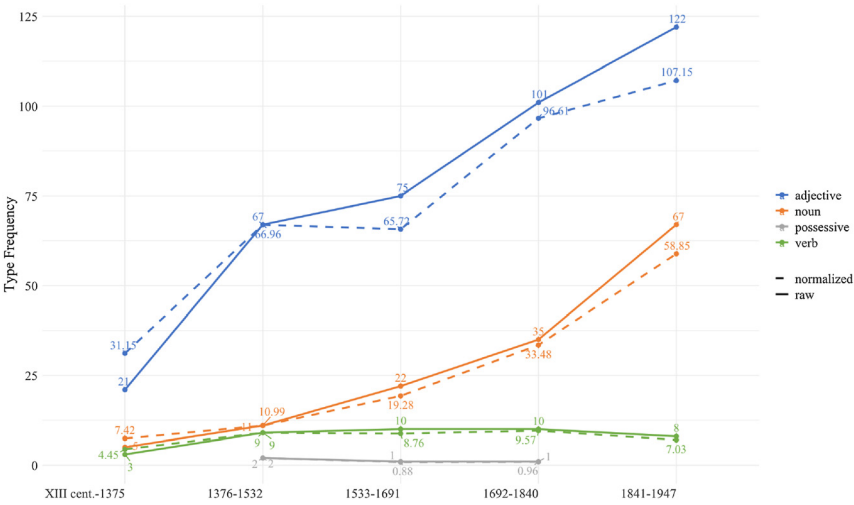


Figure 8: Type frequency (raw and normalized) of evaluative *mezzo X* across time, by part of speech.

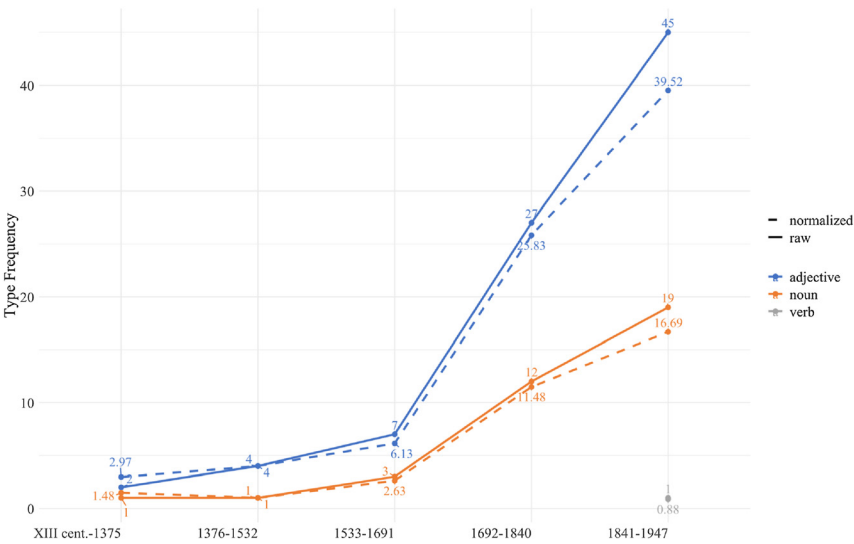


Figure 9: Type frequency (raw and normalized) of evaluative *mezzo X* across time, by part of speech.

fillers (Figure 8). This coincides with a shift from nouns referring to first-order entities (e.g., *dottore* ‘doctor’, *milite* ‘soldier’) to eventive nouns (*rifiuto* ‘refusal’, *disperazione* ‘desperation’).

The trajectory of *semi-X* is quite different. In the first period, there are only three evaluative types, two adjectival and one nominal, all of which are already found in Latin (lat. *semivivus* > it. *semivivo* ‘half-alive’, lat. *semimortuus* > it. *semimorto* ‘half-dead’, lat. *semideus* > it. *semidio* ‘demi-god’). Until the XVIII century, evaluative *semi-X* remains substantially unproductive; instead, in period 4 (1692–1840) the number of types triplicates, and then stably grows in the last period. This growth involves both nouns and adjectives, though it is most dramatic with adjectives (Figure 9). This expansion seems to coincide with a process of re-Latinization common to other prefixation strategies (Iacobini 2019, 2023). Such process, as already noted by Micheli (2023), starts from scientific and technical terminologies, and thus involves many adjectives from specialized texts (*coibente* ‘insulating’, *filosofico* ‘philosophical’, *razionalista* ‘rationalist’); nonetheless, a good number of simplex and participial adjectives appear as fillers (e.g., *nudo* ‘naked’, *aperto* ‘open’, *spento* ‘turned-off’), and their number increases in the last period.

4.2.2 Shared productivity

Crucially, alongside the increase in the productivity of the two constructions, there has also been a rise in the number of shared fillers across periods. Table 8 shows the number of fillers shared by the two constructions, while Table 9 presents the types of shared innovation (attracted and joint new types) between them.

Table 8: Number of shared fillers across periods.

Construction	XIII cent.–1375	1376–1532	1533–1691	1692–1840	1841–1947
<i>mezzo X</i>	28	89	108	145	196
<i>semi-X</i>	3	5	10	39	63
<i>n. shared fillers</i>	2	2	4	8	15

Table 9: Interactions between *semi-X* and *mezzo X* in the creation of new types, across periods.

Construction	1376–1532	1533–1691	1692–1840	1841–1947
<i>mezzo X</i>				
New types	79	86	121	155
attracted from <i>semi-</i>	0	1	0	4
joint new types	0	0	4	5
<i>semi-X</i>				
New types	2	7	34	50
attracted from <i>mezzo</i>	0	1	4	6
joint new types	0	0	4	5

What kinds of fillers are shared? Interestingly, in the first period, two of the three fillers attested with *semi-X* were already shared with *mezzo X*, namely *morto* ‘dead’ and *vivo* ‘alive’. As previously mentioned, the use of *semi-X* with these two adjectives was already found in Latin. In contrast, as far as we know, the use of *medius* or *dimidius*, both meaning ‘half’, as evaluative markers is not attested in Latin. It is therefore plausible to assume that in the early stages of the Italian language, given the low productivity of prefixation (Iacobini 2019, 2023), *mezzo X* took over the functional space occupied by *semi-X* in Latin, and in doing so “attracted” certain fillers originally used with *semi-X*, including *morto*, which is in fact the most frequently used filler with *mezzo X* in the first three periods, and in the last one.

This pattern remains stable throughout the second and third periods. In fact, the modest growth of *semi-X* in 1533–1691 involves mostly technical adjectives (e.g., *denso* ‘dense’, *meccanico* ‘mechanical’), and the increase in shared fillers is due to the attraction of *dio* ‘god’ by *mezzo X*, and *rotto* ‘broken’ by *semi-X*. By contrast, in the last two periods, along with the broader expansion of *semi-X* types, the number of shared fillers rises significantly. This is mainly due to two factors: the mutual attraction of fillers previously associated with the other construction, and the parallel creation of new types with the same filler during the same period (Table 8). The fillers attracted by *semi-X* include primarily participles (e.g., *nascosto* ‘hidden’, *sepolto* ‘buried’), but also adjectives of nationality (e.g., *italiano* ‘Italian’), and, interestingly, the only verbal filler found with *semi-X* (*desinare* ‘to dine’), even though it represents an obsolete use. On the other hand, *mezzo X* attracts from *semi-X* some adjectival fillers (e.g., *letterato* ‘literate’, *serio* ‘serious’) and a nominal one (*fischio* ‘whistle’).

However, shared fillers appear to have a stronger impact on the type-frequency distribution of *semi-X*. First, joint types and attracted fillers contribute more consistently to the expansion of *semi-X*’s productivity, making up around 22 % of its new types in both the fourth and fifth periods. Moreover, attracted fillers become *semi-X*’s leader words: in fact, the four most frequent *semi-X* types in the final period contain fillers attracted from *mezzo X*, namely *semi-nudo* ‘half-naked’, *semi-spento* ‘half-extinguished’, *semi-aperto* ‘half-open’, and *semi-chiuso* ‘half-closed’, accounting together for over 40 % of the adjectival tokens of *semi-X* in that period ($n = 82$).

Notably, this expansion of *semi-X* in the area of adjectives, traditionally associated with *mezzo X*, does not appear to have undermined the productivity of the latter. Firstly, the fillers in question remain commonly and frequently used with *mezzo X*. Moreover, *mezzo X* continues to display robust productivity with adjectives, at least on a par with *semi-X* (see Section 3).

Thus, the resulting picture is one of a positive or neutral interaction between the two constructions, driven primarily by the expanding productivity of *semi-X*. This expansion has been mediated not only by its use with technical and scientific terms, rarely found with *mezzo X*, but also through the appropriation of fillers typically used

with *mezzo X*, and, later, by the joint emergence of new co-radical types of the two constructions. Nonetheless, there does not seem to have been a shift in the lexical niche occupied by *mezzo X*. It is true that *mezzo X* shows increased use with nouns during the period of growth of *semi-X* with adjectives, and thus, it could be linked with *semi-X*'s expansion towards the use with adjectives. However, this dynamics does not seem to coincide with any significant loss of ground of *mezzo X* productivity with adjectival fillers, as testified by synchronic data.

5 Conclusions and future outlook: towards an inclusive model of interaction

Our study has investigated the relationship between two Italian evaluative HALFNESS constructions, namely *mezzo X* and *semi-X*, in order to test the notion of competition when applied to evaluative constructions. In particular, we employed productivity as an indication of the interaction dynamics between the two constructions. To this end, we performed (1) a synchronic analysis of the productivity of *semi-X* and *mezzo X* and (2) a preliminary diachronic analysis of the evolution and possible convergence in their productivity as evaluative constructions. This allowed us to check for areas of overlaps and differences in productivity, as well as the trends in productivity over time.

The results reveal that *mezzo X* and *semi-X* not only coexist, but increasingly converge in the fillers they select, getting ultimately to cover similar semantic areas in their domain of application. From the synchronic perspective, we observe that the two constructions show differences in productivity, even if such differences do not seem to be dramatic: *mezzo X* seems slightly more productive, but, when factoring out nominal fillers, which appear to be rarely used with *semi-X*, the productivity rates substantially align. Regarding the contexts of productivity, *mezzo X* seems to be used more productively in fiction prose, while, again, the productivity of the two constructions is equivalent in the Press. What is truly interesting, however, is that the two constructions seem to be potentially applicable to fillers coming from the same semantic fields, and get to be often applied to the same fillers, creating potentially synonymous doublets.

From a diachronic perspective, while *mezzo X* initially dominated as the sole productive construction, from the eighteenth century onwards *semi-X* was reintroduced into the lexicon and began generating a significant number of types with the same bases already used by *mezzo X*. The result is an increasing number of shared fillers between the two constructions, and parallel trajectories of productivity

growth, without a corresponding decrease in *mezzo X* or a shift by either construction into a new lexical niche.

The presented data, although partial, suggest that the coexistence of the two *HALFNESS* constructions is not adequately captured by a purely “exclusionist” model. In fact, the two analyses presented above have revealed some aspects that seem to challenge a strict division of labor between the two patterns:

- (1) the presence of semantic overlap and shared fillers, which results in a high number of near-synonymous forms;
- (2) an evolutionary trajectory that does not reduce the productivity of one construction in favor of the other but rather *feeds* the productivity of at least one of them (*semi-X*), eventually leading to the emergence of new types of the two constructions involving the same fillers within the same period.

The first point shows a systematic violation of the principle of isomorphism, while the second highlights a dynamic that favors the proliferation of alternative forms with the same lexical fillers, rather than constraining them. Can this situation still be reconciled with an isomorphistic approach?

It is true that competition between constructions may remain unresolved, also for extended periods. However, such an account provides little interpretive or predictive power: it could be that *semi-X* will eventually exclude or marginalize *mezzo X*, but so far, data do not provide support for such type of dynamics. Instead, what we observe is a process of attraction (De Smet et al. 2018), rather than differentiation in terms of productivity, an outcome that contradicts the assumption of a negative interaction between the two constructions.

In contrast to exclusionist interpretations, we propose that the interaction between *mezzo X* and *semi-X* could be best understood as mostly inclusive, even though we do not assume the absence of competition. Indeed, both constructions have undergone an evolution from quantitative to evaluative uses at different moments in time, and with a certain degree of independence. However, data seem to suggest that some interaction between the two constructions might have been a factor in their development. In the initial phase, the emergence of *mezzo X* with adjectival fillers seems to have been prompted by the demise of *semi-X* and, more broadly, of prefixation strategies. It remains unclear how exactly this process unfolded, and whether it reflects a form of mutualism or competition; further investigation is needed to clarify this point. However, we should note that, at least from the semantic perspective, no functional pressure towards differentiation is found. In fact, the diachronic analysis has shown that the areas of non-overlap identified in Section 3 are not due to a shift of *semi-X* into a distinct niche, but they rather reflect its original employment when reintroduced into the Italian lexicon.

What seems to have emerged in time, instead, is analogical attraction rather than differentiation. In fact, when reintroduced, *semi-X* begins to share fillers with *mezzo X*, some of which have become among its most frequent types: thus, the existence of *mezzo X* could have possibly facilitated the increased productivity of *semi-X* in non-scientific lexicon. Conversely, *mezzo X* does not appear to have been negatively or positively affected by this relationship and has continued its development independently. Thus, the interaction between the HALFNESS constructions appears to be one of inclusive coexistence, rather than competitive. Clearly, this does not preclude the possibility that they may be distinguished by virtue of some functional nuances (Author), or that division of labor exists on some other level (such as the contextual one).

We could ask ourselves whether such a relationship may be rooted in the specific nature of the constructions themselves (i.e., a neoclassical prefix and an analytic construction), in their semantic transparency and paraphrasability, or in the socio-cultural dynamics underpinning their paths. However, our findings are consistent with other studies of rivalry between evaluatives (Section 2), which consider a wide range of affixes. We therefore propose that alternative models of interaction may offer stronger explanatory power than competition when it comes to areas of grammar that are not tightly governed by isomorphism (such as word-formation). These patterns point to a relationship that goes well beyond competition: while they do not constitute direct evidence of positive interaction or facilitation, they suggest that the two constructions have grown closer over time. More broadly, the findings indicate that *mezzo X* and *semi-X* do not really behave as Darwinian rivals, as typically assumed in competition-based models. Rather, they seem to engage in a relationship of neutral or even positive interaction.

This calls into question the assumption that competition is the sole or default model for understanding interactions between quasi-synonymous constructions. Future research will aim to investigate more closely the fine-grained dynamics between *semi-X* and *mezzo X*, with the goal of uncovering whether competition may still play a role in this complex interaction. Nonetheless, we hope that the evidence presented here will contribute to broader theoretical reflection on constructional coexistence, and encourage the development of more comprehensive models of interaction, encompassing multifaceted and complex interactions between constructions.

Acknowledgments: I would like to thank the anonymous reviewers for their insightful comments and suggestions, which have significantly strengthened the article from a methodological standpoint and allowed me to improve its clarity of exposition. I am also grateful to Luisa Troncone for her feedback on an earlier version of the manuscript, and to Nicola Cirillo for training the distributional model

for contemporary Italian. The responsibility for any errors or inaccuracies is entirely mine.

Research ethics: Not applicable.

Informed consent: Not applicable.

Author contributions: The author has accepted responsibility for the entire content of this manuscript and approved its submission.

Use of Large Language Models, AI and Machine Learning Tools: Language Models (LLMs) were used in the revised version of the manuscript to assist with language editing, translation of selected passages, and improvement of code. All outputs were carefully reviewed and validated by the author.

Conflict of interest: The author states no conflict of interest.

Research funding: None declared.

Data availability: Data and code employed for performing the analyses can be found in the online repository: <https://github.com/fla-pi/beyond-competition-in-evaluation>.

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